

ZAIDH RIZME

MECHATRONICS ENGINEER

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SUMMARY

Aspiring Mechatronics Engineer with hands-on experience in embedded systems, automation, and sensor integration. Skilled in 3D CAD Modelling, C Programming, Microcontroller programming, MATLAB, with practical exposure to industrial automation through internship experience. Passionate about developing smart engineering solutions that improve efficiency and reliability.

PROFESSIONAL EXPERIENCE

Autonomation Engineer | MAS Capital - Intimates Division (Pvt) Ltd. Jul 2025 - Feb 2026

- Implemented autonomation systems to reduce downtime and defects in manufacturing lines.
- Integrated sensors, PLCs, and vision systems for process monitoring and quality control.
- Designing numerous 3D models and mechanical assemblies for automation purposes.
- Assisted in testing, troubleshooting, and documentation of autonomation solutions.

PROJECTS

ToF Based SLAM and Autonomous Mobile Robot Dec 2025 - Present

University of Wolverhampton | Final Year Project

- Development phase of a ToF-based SLAM system for autonomous localization and mapping.
- Implement sensor fusion using IMU and wheel encoder data for pose estimation.
- Designed embedded real-time navigation and motion control on ESP32.
- Developed MATLAB tools for mapping, visualization, and performance analysis.
- Integrated and validated robotic sensing, localization, and navigation subsystems.

Automated Modified Iron July 2025 - Dec 2025

MAS Capital - Intimates Division (Pvt) Ltd

- Developed an industrial iron with relay, timer, and temperature control for precise heating.
- Integrated pneumatic cylinder and movable arms to automate garment pre-heating before pressing.
- Improved efficiency, consistency, and reduced manual effort in branded apparel manufacturing.
- Conducted testing, troubleshooting, and optimization to ensure reliable production use.

Automated Oil Level Monitoring System for Industrial Machinery Oct 2024 - April 2025

Brother Industries Pvt Ltd | University of Wolverhampton

- Designed an embedded system with MS5837 sensor and ATmega328P for continuous oil level monitoring.
- Developed C firmware to process data, trigger alerts, and ensure reliable industrial performance.

Skills gained : Designing Schematics and PCBs, C Programming, Arduino C++, 3D Modelling

Wearable Active Posture Corrector Device (Research) Jan 2024 - April 2024

OREL Corporation Pvt Ltd | University of Wolverhampton

- Investigated the use of neodymium magnets and Hall effect sensors to detect upper body posture deviations with high sensitivity.

EDUCATION

Bachelor of Engineering (Hons) in Mechatronics Engineering

Sep 2023 - Sep 2026

University of Wolverhampton | CINEC Campus Pvt LTD

- Gained hands on experience in embedded systems and automation through practical labs and individual projects.

GCE Advanced Level (Physical Science Stream)

Sep 2020 - Dec 2022

Isipathana College, Colombo 05

- Combined Mathematics - B | Chemistry - B | Physics - C

CERTIFICATIONS

- Diploma in Information and Technology (Distinction)
- Sri Lankan Mathematics Olympiad (Distinction)
- Specialized in 3D CAD Modelling using Solidworks (Cousera)
- C Programming / C++ Programming at Udemy

ADDITIONAL EXPERIENCE

Academic Assistant | LivePhysics (Pvt) Ltd.

July 2024 - July 2025

- Preparing and documentation of GCE Advanced Level Physics Papers, Marking Schemes, Lesson presentations.
 - Coordinate with staff for scheduling and material planning.
- Skills gained: Excelled in MS Office documentation, Analytical Thinking, Technical Communication and Collaboration, Teamwork.

Course Instructor | NEOGRAENIUM

Jan 2023 - Sep 2024

- Developed and implemented dynamic curriculum for LEGO robotics and Micro:Bit device.

REFEREES

1. **Name:** Randeera Liyanage, Msc

Position: Lecturer in Mechatronics Engineering | Secretary - CIRS 2025 | Chairman - Industry Consultative Committee

Organization: Faculty of Engineering and Technology at CINEC Campus (Pvt) Ltd.

Email: randeera.liyanage@cinec.edu

Contact Number: +94 77 714 2029

Relationship: Lecturer | Counsellor of IEEE Student Branch CINEC Campus 2025

2. **Name:** Dr. Sanjeewa Dharmawardhana, Ph.D.

Position: Educator of Physics Subject for GCE A/L Students | Educational Consulting

Organization: LivePhysics (Pvt) Ltd.

Email: sanjeewadharmawardhana@gmail.com

Contact Number: +94 77 022 0222

Relationship: Founder | Supervisor at LivePhysics (Pvt) Ltd.